

Adaptively Grouped Contextual Bandits for Heterogeneous Human-AI Decision Making with Conformal Prediction Sets

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Personalized prediction sets with limited human feedback

AGCB learns how much AI assistance to provide by sharing feedback and refining user-task groups.

01 Why personalize the set?

A small set reduces cognitive load but may exclude the true answer. A large set preserves more options but makes human choice harder.

The best balance depends on the person and the task.

Context x	User and task features
Action α	Conformal significance level
Feedback r	Human decision quality

$$\alpha^*(x) = \arg \max_{\alpha \in \mathcal{A}} \mu(\alpha, x)$$

02 AGCB: learn, share, refine

- 1 Assign**
Route the context to its current group.
- 2 Learn**
Select α with a local bandit. Use feedback to constrain other α values.
- 3 Share**
Pool feedback across similar groups.
- 4 Refine**
Split when uncertainty falls below the group's approximation bias.

One observation can inform multiple arms

Example: under binary monotonicity, a correct choice remains correct for a smaller nested set that still covers the truth. If a set excludes the truth, all smaller sets have reward zero.

03 Structure and guarantees

Continuity enables information sharing

$$|\mu(\alpha, x) - \mu(\alpha', x')| \leq L d((\alpha, x), (\alpha', x'))^\beta$$

Conditional monotonicity

A smaller set gives no worse decision quality when it still contains the true answer. Overall expected reward need not be monotone in α .

Adaptive refinement

$$C\sigma \sqrt{\frac{\log(KT)}{n_g^{\text{eff}}}} \leq \eta L d_g^\beta$$

Split when uncertainty $\leq$ approximation bias.

Group diameter determines approximation bias. Sharing increases the effective sample size and lets well-supported regions split earlier.

Two learning objectives

S(T): suboptimality of the final policy (AGCB-SH).
R(T): loss accumulated during learning (AGCB-SE).

$$\mathbb{E}[S(T)] = \tilde{O}(T^{-\frac{\beta}{2\beta+d}}) \quad \mathbb{E}[R(T)] = \tilde{O}(T^{\frac{\beta+d}{2\beta+d}})$$

Standard Hölder contextual-bandit T-exponents.
d: context dimension. β : smoothness exponent.

Arm dependence: polynomial to logarithmic

$$K^{\frac{\beta}{2\beta+d}} \rightarrow (\log K)^{\frac{\beta}{2\beta+d}}$$

Bounds suppress L, noise, sharing factors and log terms in T.
Sub-Gaussian noise and Appendix G.1 conditions apply.

04 Evidence from human decisions

ImageNet16H-PS

194,407 predictions · 2,751 participants

21 features / 100 arms / T = 1,000 / 30 runs

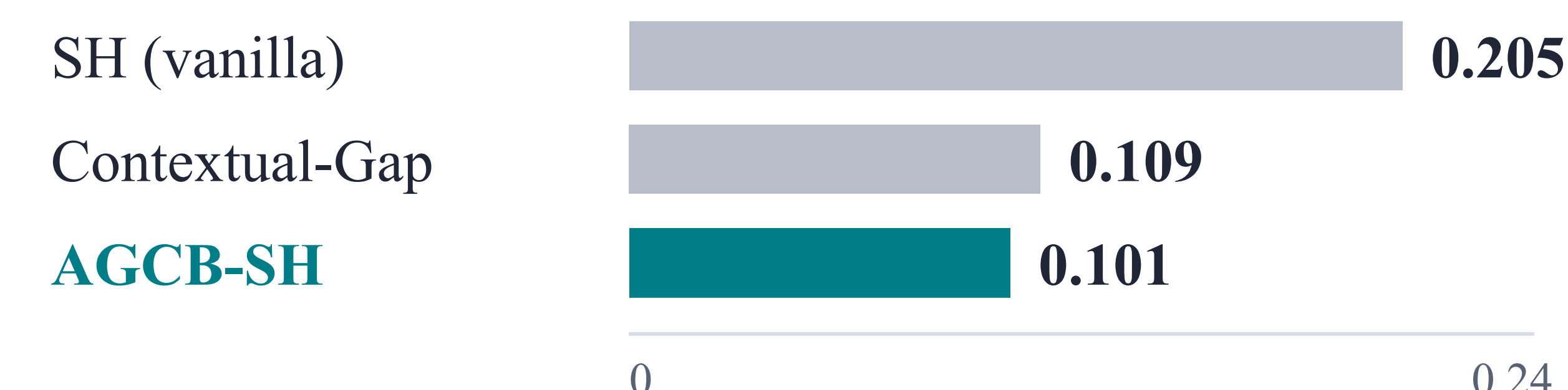
Replay uses empirical human responses. Binary feedback measures correctness. Continuous feedback adds Gaussian noise to binary rewards.

Cumulative regret · binary feedback



19.2% lower than UCB with CF

Simple regret · continuous feedback



50.7% lower than vanilla SH

Additional real-human results

Metric	AGCB	Comparison
Binary simple regret	0.103 (SH)	0.110 (vanilla SH)
Continuous cumulative	106.23 (SE)	109.86 (UCB+CF)

Means from Tables 4–5. Lower regret is better.
CF = counterfactual updates.

Synthetic (K = 100, d = 10): AGCB-SH 0.307 / 0.113 vs SH 0.421 / 0.251 (binary / continuous simple regret).

Means over 30 runs (Table 3). With 10 arms, BOED can perform better (Table 2).

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